

GeoIP Blocking using Xtables-Addons on Ubuntu

Objective

Block traffic based on country using the **xt_geoip** module in iptables.

Step 1: Install Required Packages

```
sudo apt update

sudo apt install -y \
build-essential \
gcc \
g++ \
iptables \
libxtables-dev \
linux-headers-$(uname -r) \
wget \
xz-utils \
make
```

Verify headers:

```
ls /usr/src/linux-headers-$(uname -r)
```

Step 2: Download Xtables-Addons

```
cd /tmp
```

```
wget https://inai.de/files/xtables-addons/xtables-addons-3.26.tar.xz
```

```
tar -xvf xtables-addons-3.26.tar.xz
```

```
cd xtables-addons-3.26
```

Step 3: Modify Kbuild

Open file:

```
sudo nano extensions/Kbuild
```

Comment the following lines:

```
#obj-${build_TARPIT} += xt_TARPIT.o  
#obj-${build_condition} += xt_condition.o  
#obj-${build_pknock} += pknock/  
obj-${build_quota2} += xt_quota2.o
```

Save and exit.

Step 4: Compile and Install

```
sudo ./configure
```

```
sudo make
```

```
sudo make install
```

Step 5: Load GeoIP Module

```
sudo modprobe xt_geoip
```

Verify:

```
lsmod | grep geoip
```

Expected output:

```
xt_geoip
```

Step 6: Create GeoIP Directory

```
sudo mkdir -p /usr/share/xt_geoip
```

Step 7: Download GeoIP Database

```
sudo wget -q \  
https://legacy-geoip-csv.ufficyo.com/Legacy-MaxMind-GeoIP-database.tar.gz \  
-O - | sudo tar -xvzf - -C /usr/share/xt_geoip
```

Step 8: Copy Database Files

```
cd /usr/share/xt_geoip
```

```
sudo cp LE/* .
```

```
sudo cp BE/* .
```

Verify:

```
ls *.iv4  
ls *.iv6
```

Example:

```
IN.iv4  
US.iv4  
CN.iv4  
RU.iv4  
...
```

Step 9: Verify GeoIP Support

```
sudo iptables -m geoip --help
```

You should see:

```
--src-cc  
--dst-cc
```

Step 10: Create GeoIP Rule

Block HTTPS traffic originating from US IP ranges:

```
sudo iptables -A OUTPUT \  
-m geoip \  
--src-cc US \  
-p tcp \  
-j REJECT
```

```
--dport 443 \  
-j DROP
```

Check rule:

```
sudo iptables -L OUTPUT -n -v
```

Step 11: Test

Try opening:

```
https://www.wikipedia.org  
https://www.microsoft.com
```

Observe whether connections are affected by the GeoIP rule.

Step 12: Flush OUTPUT Rules

```
sudo iptables -F OUTPUT
```

Verify:

```
sudo iptables -L OUTPUT -n -v
```

Step 13: Test Again

Open:

```
https://www.wikipedia.org  
https://www.microsoft.com
```

Websites should now be accessible normally.

Additional Examples

Block China

```
sudo iptables -A INPUT \  
-m geoip \  
--src-cc CN \  
-j DROP
```

Block Russia

```
sudo iptables -A INPUT \  
-m geoip \  
--src-cc RU \  
-j DROP
```

Allow Only India

```
sudo iptables -A INPUT \  
-m geoip \  
--src-cc IN \  
-j ACCEPT  
  
sudo iptables -A INPUT -j DROP
```

Verification Commands

```
sudo modprobe xt_geoip
```

```
lsmod | grep geoip
```

```
sudo iptables -m geoip --help
```

```
sudo iptables -L -n -v
```

Viva Questions

What is GeoIP Blocking?

Filtering network traffic based on the geographic location (country) associated with an IP address.

Which module provides GeoIP support?

```
xt_geoip
```

Command to block China?

```
iptables -A INPUT -m geoip --src-cc CN -j DROP
```

View all firewall rules?

```
iptables -L -n -v
```

Remove all OUTPUT rules?

```
iptables -F OUTPUT
```